LEARNING ACTIVITY 3

Analytical Dataset Construction

North Quay Gifts | Online Retail II

CRISP-DM Phase Data Preparation

Team or participant	Submission date
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Reading note: Text in grey italics reproduces the instructions of the original template. Text in black contains the team's results and answers, generated with the notebook "NB4 Analytical Dataset Construction - Online Retail II.ipynb".

Purpose

Transform the transactional version selected in Activity 2 into two tables with one row per customer: an RFM dataset for later segmentation and another to study repurchase in the 90 days following a cutoff.

Resources and shared rules

Use transacciones_procesadas.csv, its decision documentation, the North Quay Gifts case and Pandas. Do not repeat the cleaning. Record the selected route A or B and verify that the file keeps original dates, identifiers and movement type. The daily series from Activity 2 is reserved for later regression.

A valid purchase is a line without the C prefix, with Quantity > 0 and Price > 0, that was kept in Activity 2. A valid order is a distinct invoice with at least one of those lines. Cancellations, returns and adjustments are not new purchases by themselves. The policy keeps the recorded purchase and, in B, recognizes the credit separately on its date; it does not retrospectively rewrite that purchase.

For customer-level analyses, use only present Customer ID values and rename that column as CustomerID in the output files. Keep a consistent representation, without turning nulls into fictitious customers. Quantify the lines and amount excluded for lack of identifier. Do not remove those lines from the global files of the previous activity.

Recency and Frequency are computed with valid purchases only. Monetary and TotalSpent follow the selected route: in A, the sum of purchases; in B, purchases plus credits with their sign within the corresponding period and for the included customers. Describe the B amount as net of recorded credits. Exclude Credit movements from the purchase count.

Do not use reconciliation links found after a cutoff to retrospectively modify earlier purchases or spending. Filter the period first and aggregate afterwards; a credit only takes part if it was already recorded within that period.

Source control	Record
File, selected route and version or date	A-3-4_DataCleaning/outputs/transacciones_procesadas.csv, Route B (purchases plus credits), Activity 2 version of October 30, 2026: 1,027,016 rows × 12 columns. Checked with assert: original dates (Date = InvoiceDate without time), unique SourceRow, identifiers as text and MovementType limited to Purchase/Credit.
Available purchase and credit lines	1,007,913 Purchase lines (40,077 invoices) and 19,103 Credit lines (8,291 C invoices); period 2009-12-01 07:45:00 to 2011-12-09 12:50:00.

Source control	Record
Lines and amount excluded for lack of customer	229,201 lines (22.32%): 228,488 purchases for £3,101,456.18 and 713 credits for −£377,611.20 (net £2,723,844.98). They stay in the Activity 2 global files.
Monetary definition applied	Route B: Monetary and TotalSpent = signed sum of LineAmount of purchases plus credits of the included customers within the period (sales net of recorded credits). Recency, Frequency, TotalOrders and the label use valid purchases only. Customer totals are rounded to the penny only to remove floating-point residue.

I Building RFM for clustering

Use the whole processed transactional period, up to December 9, 2011 inclusive. Set the reference date to December 10, 2011 at 00:00, the day after the last day covered by the original file. Keep that reference even if your route does not keep a purchase on the final date.

Include identified customers with at least one valid purchase in that period. Group lines by customer and invoice when needed to distinguish orders from products; finally produce a single row per customer. Compute days with dates normalized to midnight so that the time does not subtract a day through truncation.

Output field	Definition
CustomerID	Customer identifier kept as the key
Recency	Calendar days between December 10, 2011 and the date of the last valid purchase
Frequency	Number of distinct valid-purchase invoices of the customer over the whole period
Monetary	Sum of the customer's LineAmount over the whole period, according to the documented A or B definition

Export clientes_rfm.csv with those four columns, without an extra index. Do not apply logarithms, scaling or clustering. If Monetary is zero or negative, keep it for this review and explain its origin; identify those cases before later modelling.

RFM validation control

Check	Result and evidence
Customers included and file rows	5,878 customers = 5,878 rows × 4 columns (CustomerID, Recency, Frequency, Monetary). Of the 5,939 identified customers in the file, 61 have only credits (128 lines, −£61,049.67) and do not enter.
Null and duplicate CustomerID	0 null and 0 duplicate CustomerID values.
Reference date and first and last purchase used	Reference 2011-12-10 00:00. First purchase used 2009-12-01 07:45:00; last purchase used 2011-12-09 12:50:00.
Minimum and maximum Recency and minimum Frequency	Recency min 1 and max 739 days; Frequency min 1 (max 398).
Sum of Monetary vs eligible movements	Σ Monetary £16,351,040.94 vs eligible movements £16,351,040.96 = purchases £17,374,804.27 + credits −£1,023,763.31. The £0.02 gap comes only from rounding each customer to the penny.
Customers with Monetary ≤ 0 and explanation	41 customers: 21 with exactly £0 (credits cancel all their purchases) and 20 negative (credits exceed their purchases in the file; minimum −£10,953.50, customer 12918: a £10,953.50 Manual charge credited twice). Of their 394 credit lines, 242 are reconciled with in-file purchases and 137 have no earlier purchase of the product (likely sales before Dec-2009). Kept and flagged before modelling.

1. In the notebook, manually check two customers with different numbers of orders: show their valid invoices, last date and amount according to the route. How would Frequency change if you counted lines? How do you interpret a frequent customer whose Monetary is reduced by credits?

Answer and interpretation: Customer 12378 has 2 valid invoices (509980 on 2010-05-26 and 562046 on 2011-08-02) with 301 purchase lines. Last purchase: 2011-08-02 → Recency 130. Monetary £5,416.32, with no credits. Customer 17017 has 29 valid invoices (378 lines). Last purchase: 2011-12-07 → Recency 3. Purchases £12,576.99 plus 12 credit documents (25 lines, −£6,023.20) give Monetary £6,553.79. The values computed by hand match the file.

If lines were counted, Frequency would be 301 instead of 2 for 12378 and 378 instead of 29 for 17017. On average there are 21.1 lines per order, so the median Frequency would rise from 3 to 52. Frequency would then measure basket width, not repeat visits: a customer with two large orders would look almost as frequent as one with 29.

Customer 17017 is recent and frequent (rank 152 by Frequency), but credits equal 47.9% of its purchases. Most of it comes from one manual credit (C504637, StockCode M, −£5,843.70), not from goods returned line by line. We read it as an engaged customer whose net value is lower than its gross purchases suggest. Frequency is not reduced; the low Monetary should be reviewed (adjustment or return pattern) before segmenting it as low value.

II Repurchase classification and time limits

Set the cutoff to July 1, 2011 at 00:00. The original file fully covers the required windows. Check that you use the two available years and that no prior filter has truncated the necessary dates.

Use	Start included	End excluded
180-day history	January 2, 2011 at 00:00	July 1, 2011 at 00:00
90-day repurchase observation	July 1, 2011 at 00:00	September 29, 2011 at 00:00

Select start ≤ InvoiceDate < end in both windows. Thus, the history includes all of June 30 and the label includes all of September 28. A purchase on July 1 belongs to the repurchase observation; one on September 29 is left out. The windows share no transactions.

Population and predictors

1. Build the population with identified customers who have at least one valid purchase in the history window. This list fixes the output rows. A customer who only appears after the cutoff does not enter that population.

2. Compute the following variables exclusively within the 180 historical days. In this exercise, TotalOrders and TotalSpent do not accumulate the whole history since 2009.

Field	Calculation and use
CustomerID	Customer key; kept only as an identifier
RecencyDays	Calendar days between the cutoff and the last valid historical purchase, with dates without time
TotalOrders	Distinct valid-purchase invoices in the 180 historical days
TotalSpent	Historical sum of LineAmount according to A or B, only between January 2 inclusive and the cutoff exclusive

Assignment of Repurchase90Days

3. Identify the distinct CustomerID values with at least one valid purchase in the later window. Use only positive purchases without C; a customer with only a credit in that window does not meet the condition.

4. Keep the full table of historical customers and assign 1 when the CustomerID belongs to the list of later buyers, and 0 otherwise. You may use isin() on the key or a left merge with a later list without duplicates. If you use a join, convert only the missing later flag to 0; do not fill the historical variables that way.

5. Check that the join neither drops customers nor multiplies rows. An inner join would keep only those who repurchased and would lose class 0. The absence of a later purchase means 0 within this population and this horizon; it does not imply lack of history or definitive churn.

Classification validation control

Export clientes_clasificacion.csv, without an extra index, with CustomerID, RecencyDays, TotalOrders, TotalSpent and Repurchase90Days, in that order. Keep both classes and their original proportions.

Mandatory check	Result and evidence
Final rows vs distinct customers of the historical population	2,724 rows = 2,724 distinct customers with at least one valid purchase in [2011-01-02, 2011-07-01).
Null and duplicate CustomerID	0 null and 0 duplicate CustomerID values.
Unique values of Repurchase90Days and absence of null labels	Unique values [0, 1]; 0 null labels.
RecencyDays between 1 and 180 and TotalOrders an integer of at least 1	RecencyDays from 1 to 178 (within 1–180). TotalOrders is int64, min 1 (max 67).
Minimum and maximum dates used in the predictors	2011-01-04 10:00:00 to 2011-06-30 20:08:00. There were no movements on January 2–3, 2011; the first trading day in the window is January 4.
Minimum and maximum dates used to observe repurchase	2011-07-01 08:16:00 to 2011-09-28 16:45:00 (valid purchases only).
No movements after the cutoff in TotalSpent	0 movements dated ≥ 2011-07-01 among the rows summed into TotalSpent (assert).
TotalSpent reproducible according to the route and without missing values	Σ TotalSpent £3,170,670.66 = window purchases £3,414,043.52 + window credits −£243,372.86 of the population. Independent per-customer recomputation matches; 0 missing values.

If TotalSpent is zero or negative under B, review the customer's historical purchases and credits and document the result. Do not make it positive or use future returns to correct it.

Result: 9 customers have TotalSpent ≤ 0: 5 exactly £0 and 4 negative (minimum −£1,591.20, customer 12755). For example, customer 12346 bought 74,215 jars for £77,183.60 on 2011-01-18 and the order was credited 16 minutes later, so its net is £0. Their 63 historical credits all fall inside the window: 44 are reconciled with window purchases, 14 ambiguous and 5 unreconciled. Values are kept as computed, without future returns.

Repurchase90Days	Meaning	Customers	Percentage
0	No valid purchase in the following 90 days	1,315	48.27%
1	At least one valid purchase in those 90 days	1,409	51.73%
Total	Full historical population	2,724	100%

Compute each percentage over all customers of the historical population. The sum of both classes must match the file rows and the percentages must add up to 100%, except for presentation rounding.

2. Select one customer from each class and show in the notebook the purchases that justify the label. For the customer with 0, check that it does have a valid history and no valid purchases in the later window. If it only records later credits, explain why its label remains 0.

Answer and interpretation: Class 1, customer 12347: 3 history invoices (542237 on Jan 26, 549222 on Apr 7 and 556201 on Jun 9, 2011) give RecencyDays 22, TotalOrders 3 and TotalSpent £1,494.16. In the observation window it bought invoice 562032 on 2011-08-02 (22 lines, £584.91), so the label is 1.

Class 0, customer 12406: valid history (invoice 556127 on 2011-06-09, 49 lines, £1,621.25 → RecencyDays 22, TotalOrders 1). Its only movement in the observation window is credit C562728 on 2011-08-09 (3 lines, −£134.55), with no purchase. The label stays 0 because the target measures new valid purchases, and a credit is a return, not a purchase. The credit is also not used to modify its TotalSpent, because it happened after the cutoff. 28 class-0 customers are in this situation.

3. Explain what distortion would result from including an August purchase in TotalSpent or computing RecencyDays with a purchase after the cutoff. Also check that sales to new later customers did not create additional rows.

Answer and interpretation: Both would leak the label into the predictors. 712 population customers bought in August 2011 (£544,948.08). Extending TotalSpent to September 28 would change 1,436 customers: the class-1 mean would jump from £1,782.49 to £2,972.45, while class 0 stays around £500, because by definition it has no purchases there. Computing RecencyDays with a post-cutoff purchase would give all 1,409 class-1 customers values ≤ 0 (−89 to 0), while class 0 stays at 1–178. That separates the classes perfectly with information that is not available at prediction time. Example 12347: TotalSpent £1,494.16 → £2,079.07 and RecencyDays 22 → -32.

New customers: 2,089 distinct customers bought in the observation window. 680 of them are not in the historical population (373 first-time customers and 307 older customers inactive in the history window), and 0 rows were created for them. Assigning the label with isin() keeps exactly 2,724 rows; a left merge gives the same result, while an inner join would keep only 1,409 rows and lose class 0.

4. Based on the distribution obtained, what share of the population bought again? Give a commercial reading limited to those 90 days and explain why class 0 alone does not identify definitively lost customers.

Answer and interpretation: 51.73% of the population (1,409 of 2,724) made at least one valid purchase between July 1 and September 28, 2011; 48.27% did not. Commercially, about half of the customers active in the first half of 2011 came back within one summer quarter, before the Q4 seasonal peak. The two classes are nearly balanced, so the dataset keeps both outcomes well represented.

Class 0 only means no purchase within this population and this 90-day horizon. Many customers are wholesalers with longer or seasonal buying cycles, and the window ends just before the busiest months. As a validation-only check (not used in any variable), 462 of the 1,315 class-0 customers (35.1%) bought again between September 29 and December 9, 2011. Class 0 therefore cannot be read as definitive churn.

Delivery and evaluation of the analytical datasets

Complete this control with the actual names of your files and the notebook section that reproduces them. Include a view of the first five rows of each dataset in the notebook; there is no need to copy the whole dataset into the report.

Dataset	Rows and columns	Key and period	Notebook section
clientes_rfm.csv	5,878 rows × 4 columns (CustomerID, Recency, Frequency, Monetary)	CustomerID; 2009-12-01 to 2011-12-09 inclusive, reference 2011-12-10 00:00	Section 1 (1.1 build; 1.2–1.4 checks)
clientes_clasificacion.csv	2,724 rows × 5 columns (CustomerID, RecencyDays, TotalOrders, TotalSpent, Repurchase90Days)	CustomerID; history [2011-01-02, 2011-07-01), label [2011-07-01, 2011-09-29)	Section 2 (2.2 predictors; 2.3 label; 2.4–2.7 checks)

5. Why can the two datasets include different numbers of customers even though they come from the same file? Explain the difference between the full RFM period and the historical classification population. State how the lack of identifier affected the scope of the analysis.

Answer and interpretation: RFM covers the whole period, so it includes every identified customer with at least one valid purchase between December 2009 and December 2011 (5,878). The classification population requires a valid purchase inside the 180-day history window (2,724). All of these customers are also in RFM. The 3,154 customers that appear only in RFM break down as follows: 1,659 whose last purchase was before January 2, 2011, 837 whose first purchase was after the cutoff, and 658 who bought before and after but not during the window.

Missing identifiers reduce the scope of both datasets. 229,201 lines (22.32%) with £3,101,456.18 of purchases cannot be assigned to any customer. In the history window alone, 52,169 lines and £474,989.08 (13.0% of the window's net amount) are left out. The results describe identified customers only; the global daily series from Activity 2 still includes those sales.

Deliverables

1. Executed .ipynb notebook with the construction and checks of both datasets.
2. clientes_rfm.csv.
3. clientes_clasificacion.csv.
4. This document with the controls and answers completed, as a brief validation report.

Delivered files: Notebook "NB4 Analytical Dataset Construction - Online Retail II.ipynb" (executed), outputs/clientes_rfm.csv and outputs/clientes_clasificacion.csv in A-3-5_DataConstruction, and this report. Complement: outputs/resultados_actividad3.json (figures in this report).

Activity limit

The delivery ends with the verified tables. The train/test split, stratification, transformations, algorithms and metrics will be done later. Do not build temporal covariates for regression yet.

References

Chen, D. (2012). Online Retail II [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5CG6D

North Quay Gifts Case Study. Provided material. https://docs.google.com/document/d/1luKfmlQgW3RHrsZAIXC30ydZrni3wRQvnw8s3VlLRss/edit